

An entrywise inverse-norm conjecture and its sharp replacement

Reconstructed candidate proof draft. Independent mathematical review and source/priority verification are required. The IE identifiers are provisional labels recovered from earlier notes, not verified current repository identifiers.

Abstract

A 3×3 positive strictly diagonally dominant matrix refutes the displayed entrywise inverse-norm inequality. Under the stated entrywise upper bounds, the sharp inverse infinity-norm infimum is $1/(\alpha+m)$, rather than the value at $S = \alpha I + m\mathbf{1}\mathbf{1}^T$. The conclusion concerns entrywise ordering, not a stronger order also comparing diagonal-dominance margins.

1 Counterexample

Let $n \geq 3$, $m > 0$, $\alpha \geq (n-2)m$, and $S = \alpha I + m\mathbf{1}\mathbf{1}^T$. The target as recovered asks whether a symmetric, entrywise positive, diagonally dominant $J \leq S$ entrywise must have $\|J^{-1}\|_\infty \geq \|S^{-1}\|_\infty$.

For $n = 3$, $m = \alpha = 1$, take

$$S = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad J = \begin{pmatrix} 2 & 1/2 & 1/2 \\ 1/2 & 2 & 1/2 \\ 1/2 & 1/2 & 2 \end{pmatrix}.$$

All entries of J are positive, $J \leq S$ entrywise, and J is strictly diagonally dominant. Yet

$$S^{-1} = I - \frac{1}{4}\mathbf{1}\mathbf{1}^T, \quad J^{-1} = \frac{2}{3}I - \frac{1}{9}\mathbf{1}\mathbf{1}^T, \quad \boxed{\|J^{-1}\|_\infty = \frac{7}{9} < \frac{5}{4} = \|S^{-1}\|_\infty}.$$

2 Sharp infimum

Theorem 1. *For every admissible n, m, α , over the stated class,*

$$\boxed{\inf_J \|J^{-1}\|_\infty = \frac{1}{\alpha + m}}.$$

Strict positivity of all entries prevents attainment.

Proof. All such J are positive definite, because

$$x^T J x = \sum_i \left(J_{ii} - \sum_{j \neq i} J_{ij} \right) x_i^2 + \sum_{i < j} J_{ij} (x_i + x_j)^2.$$

Every term is nonnegative, and for $n \geq 3$, vanishing of all $x_i + x_j$ forces $x = 0$. Cauchy–Schwarz applied to $J^{1/2}e_i, J^{-1/2}e_i$ gives $J_{ii}(J^{-1})_{ii} \geq 1$. Equality would require e_i to be an eigenvector, impossible with positive off-diagonal entries. With $D = \alpha + m$,

$$\|J^{-1}\|_\infty \geq (J^{-1})_{ii} > 1/J_{ii} \geq 1/D.$$

For $0 < \varepsilon \leq m$, let $J_\varepsilon = (D - \varepsilon)I + \varepsilon \mathbf{1}\mathbf{1}^T$. Its diagonal is D , its off-diagonal entries are ε , and $D \geq (n - 1)m \geq (n - 1)\varepsilon$, so it belongs to the class. Direct inversion yields

$$\|J_\varepsilon^{-1}\|_\infty = \frac{D + (2n - 3)\varepsilon}{(D - \varepsilon)(D + (n - 1)\varepsilon)} \longrightarrow \frac{1}{D}.$$

This proves both the value and nonattainment. □

The gap between the proposed value and the actual infimum is

$$\|S^{-1}\|_\infty - \frac{1}{\alpha + m} = \frac{(n - 1)m(\alpha + 2m)}{\alpha(\alpha + m)(\alpha + nm)} > 0.$$

3 Scope, checks, and sources

The exact verifier checks both inverses and the class assumptions, and tests the parameterized inverse formula at rational parameters. The universal lower bound and limiting claim are analytic arguments. A stronger ordering involving dominance margins is a different question; this example is not claimed to satisfy it.

Recovered references: OpenProblemsInNLA, provisionally IE-19, *Sharp inverse norm bound from upper bounds on matrix entries*; originating preprint arXiv:1203.6812, Conjecture 8.1.¹ The current repository statement was not retrieved. The result concerns the explicit formulation in Section 1 and must not be attributed to a different formulation without checking the source.

¹<https://arxiv.org/abs/1203.6812>. Bibliographic details and original-source conventions are recovered reference leads, not independently verified here.